A Graph Neural Network Framework for Forecasting Agricultural Search-Interest Signals in Florida

Modeling Google Trends Demand Signals as Interconnected Time Series

Dissertation Chapter — Project Framework & Implementation Plan

Target venue: International Symposium on Forecasting (IIF) / International Journal of Forecasting

Prepared June 2026

Contents

0. Overview and Research Positioning

Premise. Search-interest data from Google Trends is a noisy but timely proxy for consumer attention and latent demand. Individual crop signals are not independent: strawberries and blueberries overlap in the winter Florida season, citrus peaks against the holiday calendar, and watermelon and sweet corn track the warm-season produce cycle. This project treats each crop's weekly Trends series as a **node** in a graph, encodes the statistical and agronomic relationships among crops as **edges**, and asks whether a spatio-temporal graph neural network (ST-GNN) that exploits those relationships forecasts market interest more accurately than univariate or multivariate models that ignore the graph.

Forecasting framing (important for IIF). The International Institute of Forecasters audience rewards rigorous, reproducible benchmarking over architectural novelty alone. The contribution is therefore framed as an empirical forecasting study: a controlled comparison of graph-based vs. non-graph forecasters on a real, multivariate, seasonal dataset, with honest baselines, proper rolling-origin evaluation, statistical significance testing (Diebold–Mariano), and an ablation that isolates exactly what the graph adds. Architecture is a means, not the headline.

Goal restated. Forecast weekly market-interest signals (not sales) for a panel of Florida-relevant agricultural products, and quantify whether and how inter-crop relationships improve forecast accuracy.

1. Node Design

Recommendation: one node = one (crop × Florida) weekly Google Trends series. Each node carries a single univariate signal y_i(t): the weekly Trends interest score for crop i in the Florida geo, on a 0–100 scale. This is the cleanest, most data-feasible unit and keeps the graph interpretable (each node is a crop you can name).

1.1 The three candidate node definitions

Node definition	What it captures	Verdict for this project
Crop only (one Trends series per crop, FL geo)	Pure inter-crop structure; smallest, most stable graph (~15–25 nodes).	RECOMMENDED as the core design. Maximally feasible; every edge type below applies directly.
Crop × region (crop per FL metro/DMA)	Spatial diffusion of interest across Florida (e.g., Tampa vs. Miami).	Strong EXTENSION (adds a spatial axis, novelty). Feasible only for high-volume crops; sub-state Trends data is sparse and noisy.
Crop × platform/source (Trends + Wikipedia + social)	Cross-signal validation; multi-view demand.	Use as MULTI-VIEW node FEATURES, not separate nodes — attach Wikipedia pageviews etc. as extra input channels on the crop node.

Practical rule. Keep the node set = crops for the main study (clean, publishable, feasible). Promote crop×region to a second experiment to claim a spatial-diffusion contribution. Fold alternative platforms in as additional node features rather than multiplying nodes — this avoids a sparse, hard-to-train graph.

1.2 Node features (per node, per timestep)

Even with a univariate target, give each node a richer feature vector x_i(t). This is where seasonality and exogenous data enter:

- **Target lags:** y_i(t-1 … t-L) (e.g., L = 8–52 weeks).
- **Calendar/seasonal encodings:** week-of-year via sine/cosine (Fourier terms), month one-hot, holiday flags.
- **Crop-calendar flags:** binary in-season / harvest indicator for crop i from UF/IFAS and USDA crop calendars.
- **Exogenous (optional channels):** Florida weekly temperature/precip, Wikipedia pageviews for the crop, USDA price or shipment indices where available.
- **Static node attributes:** crop family, typical season (winter/summer/perennial), perishability — used by attention models and for community analysis.

2. Edge Design

Edges encode "which crops' interest move together / lead / lag." Define **several candidate adjacency matrices**, treat the edge definition as a first-class experimental factor, and compare them. Build each on the **training split only** to avoid look-ahead leakage.

2.1 Candidate edge definitions

Edge type	Definition	Notes / when it helps
Pearson correlation	Edge weight = corr(y_i, y_j) on (log/differenced) series; threshold |r| ≥ τ.	Cheap, interpretable baseline. Use on stationarized series so shared trend/seasonality doesn't inflate r.
Spearman correlation	Rank correlation; robust to outliers & monotone nonlinearity.	Good default for spiky Trends data; report alongside Pearson.
Lagged / cross-correlation	Max cross-correlation over lags k∈[1,K]; store the argmax lag → directed edge i→j.	Directly tests 'does crop i lead crop j?'. Produces a DIRECTED graph — core to your strawberry→blueberry question.
Dynamic Time Warping (DTW)	Edge = 1/(1+DTW distance) on z-normalized series.	Captures shape similarity under phase shifts (seasonal peaks of differing width). Pair with a window constraint (Sakoe–Chiba).
Seasonal similarity	Correlation of seasonal components (STL decomposition) or of week-of-year climatology profiles.	Isolates 'same season' structure from short-term co-movement; underpins the community-detection RQ.

Edge type	Definition	Notes / when it helps
Co-search / semantic	Trends 'related queries', co-occurrence, or embedding cosine similarity (crop name embeddings / co-purchase).	Adds behavioral/semantic relatedness Trends co-movement misses (e.g., recipe-driven pairings).
Agronomic / domain graph	Expert edges: shared growing season, region, crop family, substitute/complement in consumption.	Knowledge-based prior from UF/IFAS & USDA; strong, defensible, and great for interpretation.
Learned / adaptive	Self-adaptive adjacency learned end-to-end (Graph WaveNet style) or attention weights (GAT).	Lets the model discover edges; compare what it learns against the agronomic graph as validation.

2.2 How to use multiple edge sets

1. **Single-graph experiments:** train the same ST-GNN under each adjacency (Pearson, Spearman, DTW, lagged, seasonal, semantic, agronomic) and compare accuracy — this answers 'which relationship matters'.
2. **Fused graph:** combine via weighted sum / multiplex stacking, or feed several relations to a relational/multi-relational GNN (R-GCN-style).
3. **Learned vs. fixed:** compare a learned adaptive adjacency against the best fixed one; inspect whether the learned graph rediscovers agronomic communities.

Graph construction hygiene. Stationarize before correlation (log + seasonal differencing); build adjacency on training data only; sparsify (k-NN per node, e.g., k=3–5, or threshold) so the GNN sees structure, not a near-complete graph; row-normalize weights. Report graph density and degree distribution.

3. Recommended Florida Agricultural Products

Choose crops that (a) are economically important in Florida, (b) have enough search volume for a usable Trends signal, and (c) span complementary seasonal cycles so the graph has real structure. Florida's signature is its **winter/early-spring fresh-produce window** (when most of the U.S. cannot grow), plus perennial citrus and a large nursery/greenhouse sector.

3.1 Core panel (≈15–18 nodes) — recommended starting set

Crop	FL season	Why include	Trends term tip
Strawberries	Winter–Spring (Dec–Mar)	Iconic FL winter crop (Plant City); your existing series.	"strawberry"
Blueberries	Spring (Mar–May)	Early-market premium crop; tests strawberry→blueberry lead/lag.	"blueberry"

Crop	FL season	Why include	Trends term tip
Oranges / Citrus	Fall–Winter (Oct–Jun)	FL's defining perennial; holiday & juice demand.	"oranges", "citrus"
Grapefruit	Winter (Nov–May)	Distinct citrus signal; Indian River brand.	"grapefruit"
Tomatoes	Fall & Spring (Oct–Jun)	Largest-value FL vegetable; long window.	"tomatoes"
Watermelon	Spring–Summer (Apr–Jul)	Strong summer seasonality; contrasts winter crops.	"watermelon"
Sweet corn	Fall & Spring	Major FL vegetable; warm-season pattern.	"sweet corn" / "corn"
Bell peppers	Fall–Spring	High-value vegetable; season overlaps tomato.	"bell pepper"
Cucumbers	Fall & Spring	Companion vegetable signal.	"cucumber"
Snap / green beans	Fall–Spring	Broad fresh-market vegetable.	"green beans"
Peanuts	Summer–Fall (Apr plant–Oct)	Major FL field crop; distinct harvest cycle.	"peanuts"
Avocados	Summer–Winter (Jun–Mar)	S. Florida tropical; long, off-phase season.	"avocado"
Mangoes	Summer (May–Sep)	Tropical S. FL; strong summer peak.	"mango"
Squash / zucchini	Fall–Spring	Vegetable-cluster member.	"squash" / "zucchini"
Sugarcane	Fall–Spring harvest	Largest FL acreage field crop (EAA).	"sugarcane"
Nursery / ornamental plants	Year-round, spring peak	FL's #1 ag sector by value; garden-season demand.	"plant nursery" / specific plants

3.2 Optional extension crops

Add as data volume allows: **potatoes, cabbage, lettuce, eggplant, okra, sweet potatoes, peaches (FL low-chill), guava, lychee, papaya, honey, cattle/beef,** and **hemp**. For low-volume tropicals, expect noisy Trends signals — verify each term returns a non-flat series before committing it as a node.

Term-selection caution. Prefer Trends Topic entities over raw search strings where possible (disambiguates 'orange' the fruit vs. the color/county/phone brand). Validate every term's series for volume and meaning before locking the node set.

4. Data Collection Plan

Current data-access reality (as of mid-2026). The widely used **pytrends** library was archived (read-only) in April 2025 and its endpoints are unreliable. Google launched an **official Google Trends API in July 2025, but it remains in limited alpha** (waitlist access; ~5-year/1800-day lookback; daily/weekly/monthly aggregation; region and sub-region geo). Plan around this rather than assuming pytrends works.

4.1 Recommended acquisition strategy (in priority order)

4. **Apply for the official Google Trends API alpha now.** It is the cleanest long-term source, gives consistently scaled weekly data with sub-region geo, and is the most defensible 'data and methods' provenance for a journal. Lead times are long, so apply early.
5. **Manual / semi-automated Trends CSV export as the dependable fallback.** The trends.google.com UI lets you export interest-over-time CSVs per term for geo=US-FL. This is what you already did (your strawberry CSV). Script the URL/UI flow, or use the browser to pull each crop once, then refresh periodically.
6. **A maintained third-party API if you need scale/automation.** SerpApi, ScrapingBee, Apify, or Glimpse expose stable Trends endpoints (paid). Useful for many terms × geos and for reproducible refreshes; cite the provider and access date.
7. **Wikipedia pageviews as a free, fully-open companion signal.** The Wikimedia Pageviews API is stable and unrestricted — excellent as an exogenous node feature and as a robustness cross-check on Trends.

4.2 The weekly-vs-monthly scaling problem (handle explicitly)

Google Trends returns **weekly** data only when the requested window is roughly < 5 years; longer windows return **monthly** data, each request is rescaled to its own 0–100 max, and separate requests are not directly comparable. Your current files are monthly (2020→), so to get a multi-year WEEKLY panel you must stitch overlapping windows.

Stitching recipe:

8. Pull weekly data in overlapping windows (e.g., rolling ~4.5-year requests, or 6–12 month windows with overlap).
9. Pull one long MONTHLY series as a 'backbone' for each crop.
10. Rescale each weekly window onto the monthly backbone using the overlap ratio (or use a published library that reconstructs continuous Trends series).
11. To keep multiple crops on a COMMON scale, request them together in the same multi-term query (Trends scales them jointly, max=100 across the group), then anchor groups via a shared reference term.

4.3 Geography: statewide vs. metro

Use statewide Florida (geo = US-FL) for the main study. It is the only level with reliable weekly volume across all crops, and it matches the dissertation's framing. Reserve **metro/DMA-level** (e.g., Miami, Tampa, Orlando, Jacksonville) for the crop×region spatial-diffusion EXTENSION, and only for high-volume crops where the sub-state series is not mostly zeros. Always record geo code, term, window, and pull date for reproducibility.

4.4 Target panel specification

Field	Specification
Frequency	Weekly (stitched), with monthly backbone retained
Span	≥ 5 years (e.g., 2019–2026) to capture ≥4 full seasonal cycles
Geo	US-FL statewide (primary); selected FL DMAs (extension)
Nodes	15–18 crops (core) → up to ~25 (extended)
Per-node features	lags, Fourier calendar terms, in-season flag, optional weather/Wikipedia/USDA
Exogenous	NOAA/PRISM FL weather; USDA-NASS / AMS; UF-IFAS crop calendars; Wikimedia pageviews

5. Graph Forecasting Approach

Combine a **graph operator** (mixes information across related crops) with a **temporal operator** (models each series over time). Implement in **PyTorch Geometric / PyG-Temporal**; build/inspect graphs in **NetworkX**.

Model	Mechanism	Role in your study
GCN + LSTM/GRU	Spectral graph conv per step → recurrent over time.	Simple, transparent first ST-GNN; good headline graph model.
GAT + GRU	Attention-weighted neighbor aggregation + GRU.	Learns edge importance → interpretable; compare attention vs. agronomic edges.
STGCN	Fully convolutional spatio-temporal (gated temporal conv + graph conv).	Fast, strong, parallel; excellent benchmark backbone.
DCRNN	Diffusion convolution on a DIRECTED graph + GRU (encoder–decoder).	Best fit for LAGGED/directed edges (lead–lag); directly tests directionality.
Graph WaveNet	Dilated causal TCN + self-adaptive learned adjacency.	Learns the graph end-to-end → discover relations; compare learned vs. fixed graph.
Temporal Graph Networks (TGN)	Memory modules over time-stamped graph events.	For dynamic/evolving edges (relationships change by season) — advanced extension.

Suggested model progression. Start with STGCN and DCRNN (mature, well-benchmarked), add Graph WaveNet for the learned-graph story, and use GAT for interpretability. Keep one consistent forecasting setup (horizon, input window, scaling) across all so comparisons are clean.

6. Research Questions and Hypotheses

Frame each RQ with a testable hypothesis and the experiment that answers it.

Research question	How it is tested
RQ1 — Do inter-crop relationships improve forecast accuracy?	Graph models vs. univariate/multivariate non-graph baselines; DM test on errors. (Primary contribution.)
RQ2 — Which relationship type matters most?	Same model under Pearson / Spearman / DTW / lagged / seasonal / semantic / agronomic adjacencies; rank by accuracy.
RQ3 — Do seasonal crops form communities?	Community detection (Louvain/Leiden) on the graph; test alignment with crop-calendar seasons.
RQ4 — Does strawberry interest predict blueberry/citrus interest?	Directed lead–lag edges + Granger-style causality + DCRNN ablation removing the edge.
RQ5 — Are lagged (directed) relationships useful?	Directed vs. undirected graph; cross-correlation lag features; accuracy delta.
RQ6 — Does a learned graph rediscover agronomic structure?	Compare Graph WaveNet's adaptive adjacency to the expert agronomic graph (edge overlap, modularity).
RQ7 — Do exogenous signals (weather, crop calendar, Wikipedia) add value on top of the graph?	Feature-ablation across node-feature sets.

7. Experiments and Evaluation

7.1 Baselines (must include honest, strong ones)

- **Naïve / seasonal-naïve:** y(t)=y(t-1) and y(t)=y(t-52). The seasonal-naïve is a HARD baseline for seasonal data — beating it is the real bar.
- **Classical:** ARIMA / SARIMA, ETS, Theta, and Prophet (per series).
- **Multivariate non-graph:** VAR (uses cross-series info WITHOUT a graph — key control for RQ1), plus a global LSTM/GRU and a Transformer (e.g., Informer/PatchTST-style) trained across all series.
- **Graph models:** STGCN, DCRNN, Graph WaveNet, GAT+GRU, GCN+LSTM.

The decisive comparison for RQ1 is GNN vs. VAR/global-Transformer — both see all series; only the GNN sees the explicit graph. That isolates the value of structure, not just of multivariate information.

7.2 Evaluation protocol

- **Rolling-origin (expanding-window) backtesting** with multiple folds; report mean ± std across folds. No shuffling; strict temporal splits.
- **Horizons:** h = 1, 4, 8, 12 weeks (short vs. seasonal-range).
- **Graphs rebuilt on each fold's training data only** (no leakage).
- **Significance:** Diebold–Mariano tests between model pairs; Friedman + Nemenyi across many series for ranking (IIF-standard).

7.3 Metrics

Metric	Why / note
MAE	Robust scale-dependent error; primary loss.
RMSE	Penalizes large misses (interest spikes).
MAPE / sMAPE	Scale-free, comparable across crops; sMAPE preferred (MAPE unstable near 0-interest weeks).
MASE	Scaled vs. seasonal-naïve — the IIF-favored metric; <1 means you beat the naïve.
Correlation (Pearson/Spearman of forecast vs. actual)	Directional/co-movement quality of the signal.
Pinball loss / coverage (optional)	If you add probabilistic forecasts (a publishable extension).

8. Novelty and Extensions (Toward Publication)

- **Edge-definition as the contribution:** a systematic study of how graph construction (correlation vs. DTW vs. lagged vs. agronomic vs. learned) affects forecasting — a clean, IIF-friendly empirical story.
- **Domain-informed graph:** fuse UF/IFAS crop calendars and USDA regional/seasonal data into an expert agronomic adjacency; show domain priors beat purely statistical graphs (or that learned graphs recover them).
- **Exogenous fusion:** Florida weather (NOAA/PRISM), USDA-NASS/AMS shipments & prices, Wikipedia pageviews, and (optionally) social signals as node features — quantify marginal value.
- **Spatial extension:** crop×DMA nodes to study geographic diffusion of interest across Florida metros.
- **Dynamic graphs:** season-dependent adjacency (edges that switch on/off by crop calendar) via TGN — relationships that evolve through the year.
- **Probabilistic forecasting:** prediction intervals + pinball loss; very well received at IIF.
- **Interpretability:** attention/learned-adjacency analysis + community detection that maps onto real agronomic seasons — turns a black box into an agricultural-economics narrative.
- **External validity:** relate Trends-based interest to USDA shipment/price movements to argue the signal has real economic meaning (nowcasting angle).

9. End-to-End Modeling Pipeline

The pipeline has six stages. Pseudocode for each follows.

9.1 Stage A — Data acquisition & weekly stitching

```
# A. Acquire weekly Florida Trends panel (stitched to common scale)
crops = ["strawberry","blueberry","oranges","grapefruit","tomatoes",
         "watermelon","sweet corn","bell pepper","cucumber","peanuts",
         "avocado","mango","squash","sugarcane","plant nursery"]
```

```
for term in crops:
    weekly  = fetch_trends(term, geo="US-FL", window=overlapping_5yr_windows)
    monthly = fetch_trends(term, geo="US-FL", window=full_span)    # backbone
    series[term] = rescale_weekly_to_monthly(weekly, monthly)     # stitch

# joint query in shared groups -> common 0..100 scale across crops
panel = anchor_groups_to_reference(series, reference_term="tomatoes")
panel = panel.asfreq("W").interpolate()        # uniform weekly index
save(panel, "fl_weekly_panel.parquet")         # T x N matrix
```

9.2 Stage B — Feature engineering

```
# B. Build per-node feature tensor X[t, node, feature]
X = {}
for i in nodes:
    y = panel[i]
    feats = [lag(y, k) for k in 1..L]                        # autoregressive
    feats += fourier_terms(week_of_year, K=3)               # seasonality
    feats += [in_season_flag[i], holiday_flag]              # crop calendar
    feats += [fl_temp, fl_precip, wiki_pageviews[i]]        # exogenous (opt.)
    X[i] = standardize(stack(feats))                        # fit scaler on TRAIN only
```

9.3 Stage C — Graph construction (per training fold)

```
# C. Build candidate adjacency matrices on TRAIN split only
S = stationarize(panel_train)        # log + seasonal differencing

A_pearson  = threshold(corr(S, method="pearson"),  tau)
A_spearman = threshold(corr(S, method="spearman"), tau)
A_dtw      = kNN(1/(1+dtw_matrix(znorm(S), window=w)), k=4)
A_lag, lag = directed_max_xcorr(S, max_lag=K)    # i->j with best lag
A_season   = corr(stl_seasonal(panel_train))     # seasonal-component corr
A_semantic = cosine(name_embeddings)             # or related-query overlap
A_agro     = expert_graph(UF_IFAS_calendar, USDA_region, crop_family)

A_set = {pearson, spearman, dtw, lag, season, semantic, agro, fused}
for A in A_set: A = row_normalize(sparsify(A))
G = networkx.from_numpy_array(A)   # inspect density, degree, communities
```

9.4 Stage D — Model definition (ST-GNN, schematic)

```
# D. Spatio-temporal GNN (PyG-Temporal); same skeleton for STGCN/DCRNN/...
class STGNN(Module):
    def __init__(self, n_feat, hidden, A):
        self.graph_op = GraphConv(n_feat, hidden, A)   # GCN/GAT/diffusion
        self.temporal = GRU(hidden, hidden)            # or TCN / LSTM
        self.head     = Linear(hidden, horizon)
    def forward(self, X_window):           # X_window: [batch, T, N, F]
        h = [self.graph_op(X_window[:,t]) for t in range(T)]
        h = self.temporal(stack(h))        # mix across time
        return self.head(h[-1])            # [batch, N, horizon]
```

9.5 Stage E — Training & rolling-origin evaluation

```
# E. Rolling-origin backtest, leakage-safe
for fold in expanding_window_folds(panel):
```

```
    train, val, test = fold
    A = build_graph(train)                  # rebuild per fold
    scaler = fit(train); apply(scaler, val, test)
    model = STGNN(...); train_with_early_stop(model, train, val, loss=MAE)
    for h in [1,4,8,12]:
        yhat = model.predict(test, horizon=h)
        log_metrics(MAE, RMSE, sMAPE, MASE, corr, fold, h)
```

9.6 Stage F — Benchmarking, tests & interpretation

```
# F. Compare, test significance, interpret
results = collect(all_models x all_graphs x folds x horizons)
rank_table   = aggregate(results, by=["model","graph"])   # mean +/- std
dm_pvalues   = diebold_mariano(best_gnn, [seasonal_naive, ARIMA, VAR, Transformer])
cd_diagram   = friedman_nemenyi(results)                  # IIF-style ranking
communities  = louvain(best_graph); compare_to(crop_calendar_seasons)
attn_or_adj  = inspect(model.learned_adjacency or attention)
```

10. Graph Schema (Summary)

Element	Specification
Graph G = (V, E)	Directed, weighted (use undirected variants for correlation/DTW graphs).
Node v_i	One Florida crop; carries weekly target y_i(t) + feature vector x_i(t).
|V|	15–18 (core) → ~25 (extended); + crop×DMA in spatial extension.
Node features	Lags, Fourier calendar terms, in-season flag, weather, Wikipedia, USDA (optional).
Edge e_ij	Relationship weight; one of {Pearson, Spearman, DTW, lagged-directed, seasonal, semantic, agronomic, learned}.
Edge weight	Normalized similarity / |corr| / 1/(1+dist); sparsified (k-NN or threshold), row-normalized.
Targets	Weekly interest y_i(t); multi-horizon h ∈ {1,4,8,12}.
Splits	Rolling-origin; graph + scalers fit on training portion only.

11. Implementation Roadmap (Next Steps)

12. **Apply for the Google Trends API alpha** today; in parallel script the manual FL CSV export for all core crops.
13. **Extend your existing strawberry pipeline** to the full 15–18-crop weekly panel; solve the weekly-stitching/scaling step and save a clean T×N parquet.
14. **Build the candidate graphs in NetworkX** (Pearson, Spearman, DTW, lagged, seasonal, agronomic); visualize and sanity-check communities against the crop calendar.
15. **Stand up baselines first** (seasonal-naïve, SARIMA, Prophet, VAR, global LSTM, Transformer) with the rolling-origin harness and metric logging.
16. **Implement STGCN and DCRNN** in PyG-Temporal; confirm they beat seasonal-naïve and VAR before adding complexity.

17. **Run the edge-type ablation and learned-graph (Graph WaveNet)** experiments; add DM tests and CD diagrams.
18. **Layer extensions** (exogenous fusion, spatial DMA nodes, dynamic graphs, probabilistic intervals) as space/novelty allows.
19. **Write up** with reproducibility appendix (data pull dates, geo codes, graph configs, seeds) — essential for IIF/IJF.

Tooling stack: Python · pandas · statsmodels/pmdarima · Prophet · NeuralForecast/Darts (baselines & Transformers) · NetworkX (graphs) · PyTorch + PyTorch Geometric + PyG-Temporal (ST-GNNs) · dtaidistance/tslearn (DTW) · Wikimedia & (alpha) Google Trends APIs.

12. Notes on Data-Access Sources

Key references consulted on current Trends data access (verify access terms before relying on any paid provider):

- Google: Introducing the Google Trends API (alpha), July 2025
- Search Engine Land — Google launches Google Trends API
- ScrapingBee — Best Google Trends APIs for 2026 (pytrends archived; provider comparison)
- SerpApi — Google Trends API in Python (pytrends alternative)
- Wikimedia Pageviews API (free exogenous signal)
- UF/IFAS Florida crop calendars & USDA-NASS / USDA-AMS specialty-crop shipments — for agronomic edges and exogenous features.